

Coin Tray - Project Brief

Local-first coin identification & pricing

Desktop + Android companion . privacy-preserving numismatic workflow

Problem

Sorting a mixed box of coins needs a quick identity and a rough market range without uploading every photo to a cloud AI service or paying per-token vision APIs.

Solution

Coin Tray runs Ollama vision models on your PC, lets you correct fields while holding the coin, stamps structured catalog baggie IDs, then prices from an allowlist of guides and marketplaces with citations. An optional Android CameraX app captures on the phone and talks to the PC over home Wi-Fi.

Architecture (one line)

Webcam / Phone camera ? PC engine (Ollama VL) ? Edit + Catalog ID ? Allowlisted pricing ? Tray (JSON / CSV)

Android is a thin client: `http://LAN:8722 + X-Coin-Tray-Token`. Ollama never leaves the PC.

Resume bullets

- * Built a local-first coin sorting system with on-machine vision ID (no cloud AI billing).
- * Implemented allowlisted live pricing (guides, auctions, marketplaces) with IQR ranges, sold vs asking, and source citations.
- * Designed catalog baggie IDs ('CA-0007 . U . 1964 25c Ag') plus disposition workflow (Keep / Sell / Check / Face).
- * Shipped a Kotlin / Jetpack Compose / CameraX Android companion over a LAN HTTP API.
- * Hardened the companion path for home use: header-only token auth, no internet exposure, collection data kept out of git.

Stack

Desktop - Python, Tk, OpenCV

Engine - 'coin_engine.py', Ollama VL + text

Phone API - stdlib 'ThreadingHTTPServer'

Android - Kotlin, Compose, CameraX

Requirements

- * Ollama + a vision model (e.g. 'qwen3-vl:8b' or 'gemma3:4b')
- * Python 3 + 'pip install -r requirements.txt'
- * Optional: Android Studio (JDK 17/21), phone on same Wi-Fi

Status

Personal / portfolio project. Estimates are comps, not appraisals. Not a substitute for professional grading.

Links

- * Repository: <https://github.com/thestickiestsyrop/PublicCoinCollector>
- * Docs: 'README.md', 'docs/ARCHITECTURE.md', 'docs/OPERATOR.md'
- * License: MIT